# C++ Programming

Juha Yli-Hemminki

07.11.2025

SeAMK

SeAMK

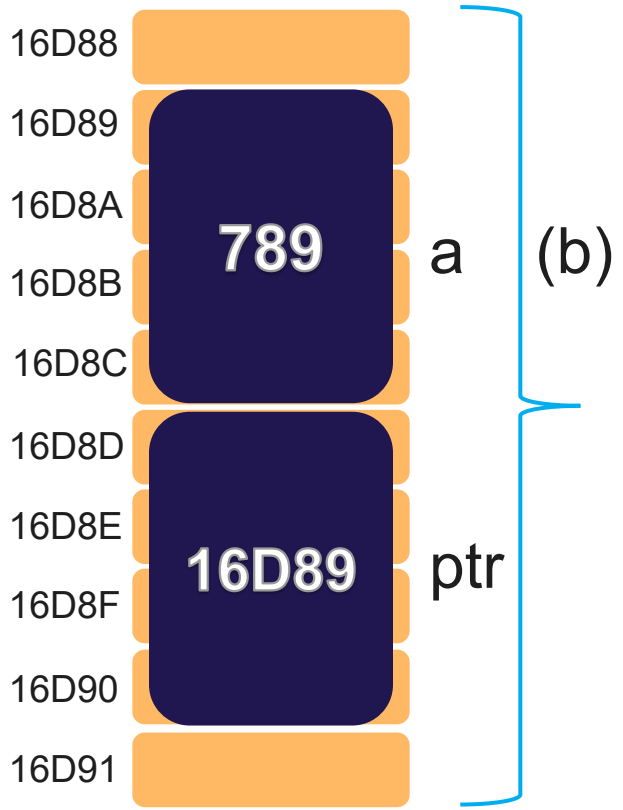
# Computer memory

Memory address

```
int a = 789;

Reference:
int& b = a; // alias for a

Pointer:
int* ptr = &a; // gets the value 16D89
```




Computer memory

SeAMK

# Computer memory

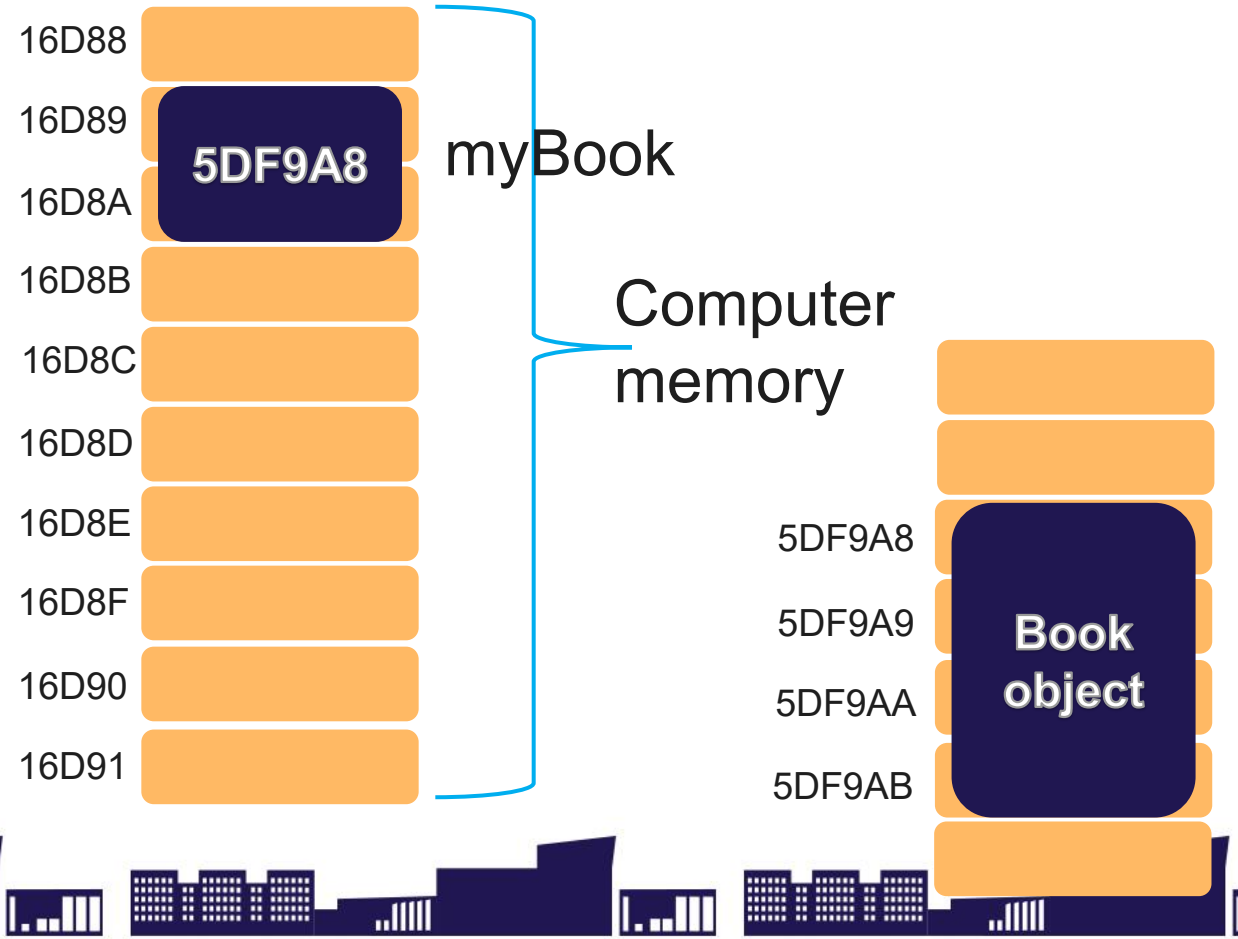
Memory address

Pointer:

Book* myBook;

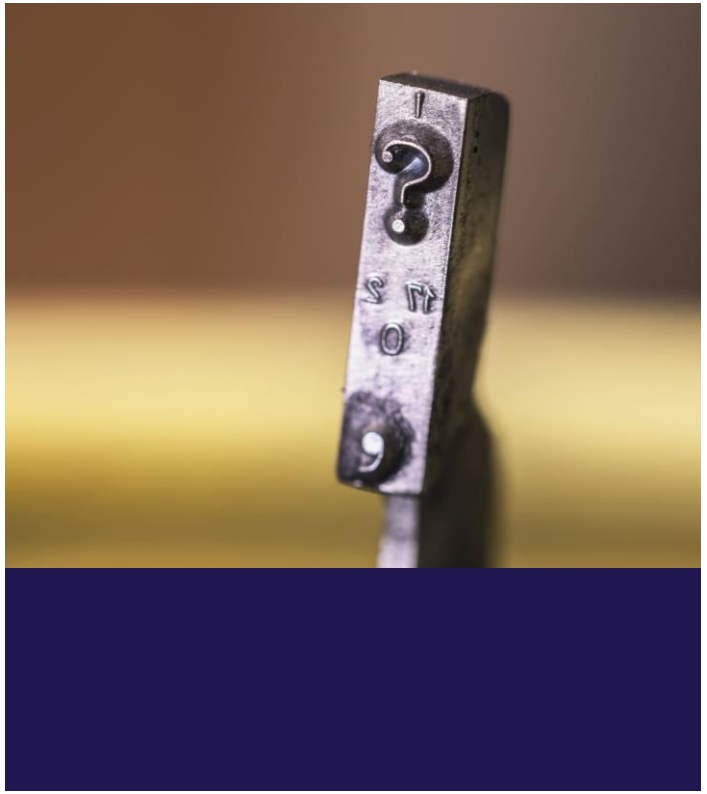
Creating new object returns the memory address:

myBook = new Book();



# Class vs Object

- **Objects** are created from **classes** during program execution
- The operation of the program is built on the cooperation of objects
- Objects may receive action requests(methods called) and request actions (call methods) from other objects
- Object may change its member values by the methods called from outside



SeAMK

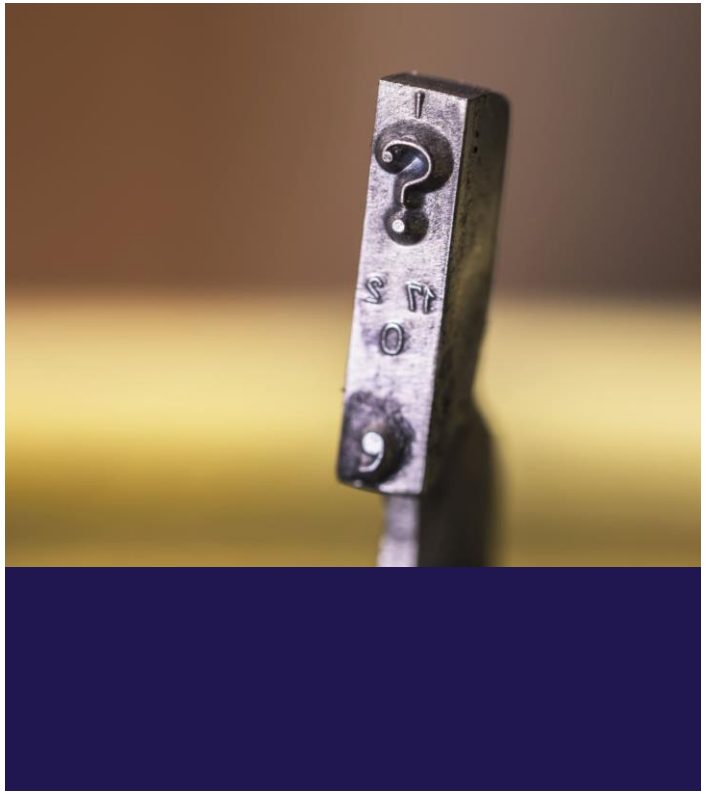
# Class vs Object

- It may be considered that a class serves as a model by which objects are made
- Sometimes a class is compared to a stamp and an object is a stamp made with a stamp, respectively
- Class is written into the code and object is created when program is run

SeAMK

# Class vs Object

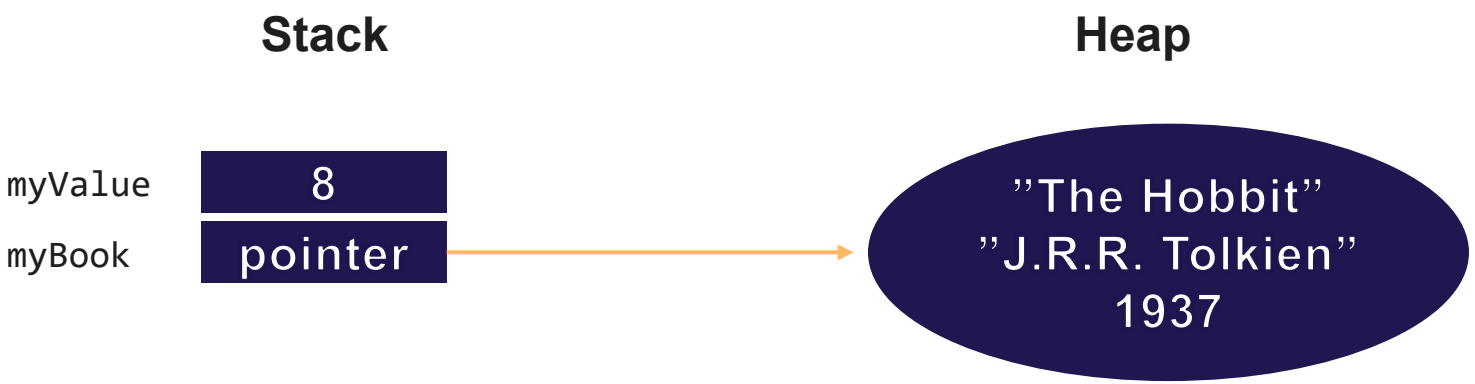
- Variables and functions are defined in the class, which become properties of the object created from it
- A class written in program code is used only through objects made from it
- If no object is created from the class written in the code, it is not used in the program at all



SeAMK

# Stack and Heap

```
int myValue = 8;
Book* myBook = new Book(”The Hobbit", "J.R.R. Tolkien”, 1937);
```



SeAMK

# Binary numbers

0000 0001 = 1
0000 0010 = 2
0000 0100 = 4
0000 1000 = 8
0001 0000 = 16
0010 0000 = 32
0100 0000 = 64
1000 0000 = 128

0000 0011 = 3
0000 0101 = 5
0000 0110 = 6
0000 0111 = 7
1111 1111 = 255

# Bitwise operators

- & (bitwise AND) result is 1 only if both bits are 1
- | (bitwise OR) result is 1 if any of the two bits is 1
- ^ (bitwise XOR) result is 1 if two bits are different
- ~ (bitwise NOT) inverts all the bits of the number
- >> (right shift) shifts bits right by given number of steps
- << (left shift) shifts bits left by given number of steps

SeAMK

SEINÄJOEN AMMATTIKORKEAKOULU

SEINÄJOKI UNIVERSITY OF APPLIED SCIENCES